PREDICTING ATP SINGLES MATCH OUTCOMES WITH SUPERVISED LEARNING: BEATING ELO BUT NOT THE MARKET

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ABSTRACT

We predict the winner of ATP singles matches (2000–2024, 65,281 usable matches) as binary classification under a symmetric representation in which the two players are assigned to slots A and B at random. All 38 features are built causally in a single chronological pass in which history is read before it is updated. Eleven classifiers are compared with rolling-origin cross-validation over 32 quarterly folds, through paired differences reported with 95% confidence intervals. The five best (two penalised logistic regressions, a linear SVM and two tree ensembles) are indistinguishable at 0.577 log-loss, and all beat a tuned surface-blended Elo by 0.0093 (95% CI $[0.0068, 0.0117]$). On an untouched 2019–2024 hold-out the model improves on Elo by 1.4 accuracy points but loses to the betting market by 2.2, in all 29 segments examined, with the gap widening almost fivefold where player history is thin. The binding constraint is the information available in result data, not the choice of classifier.

1 Introduction

Forecasting a tennis match is an unusually well-posed prediction problem: the outcome is binary and unambiguous, the covariates are public, and an external, financially incentivised forecaster publishes its own probabilities as bookmaker odds. That last property makes the exercise falsifiable. A model is useful not because it reaches some conventional accuracy, but because it carries information a market with money at stake does not already have.

Two failure modes dominate, and neither produces an error message. *Label ordering*: in the standard data format the winner is always recorded first, so a feature computed as "winner minus loser" encodes the answer outright. *Temporal leakage*: nearly every attribute of a player is derived from past matches, and a careless pipeline lets information flow backwards in time. Much published work reports a single accuracy figure on one split, without statistical comparison across models, without probabilistic metrics, and (when odds are used) treating the bookmaker as a data source rather than a competitor. Our contribution is not a new classifier but a measurement of where this problem's ceiling lies, and of which choices actually move the result:

- a causal feature pipeline whose leak-freeness is tested, not asserted;
- three negative results: head-to-head history contributes nothing, injury proxies contribute nothing, and the market cannot be beaten.

2 Related Work

Tennis forecasting divides into three families. *Point-based hierarchical models* estimate serve and return-winning probabilities and propagate them through the scoring system [1]: interpretable, but requiring point-level inputs and assuming independence between points. *Rating systems* summarise a player in a scalar updated after each match;

Elo [2] remains the strongest single-number predictor, and in a systematic comparison Kovalchik [3] found Elo-based forecasts superior to both regression and point-based models, with later refinements weighting the update by tournament importance or surface [4]. *Machine-learning approaches* build feature vectors from match history and apply general-purpose classifiers [5]. A separate literature treats odds as probability forecasts and addresses removal of the overround [6]; we compare against the raw implied probability, since that margin is exactly what a profitable strategy must overcome. Methodologically we follow Demšar [7] on comparing classifiers across resampling folds and Bergmeir and Benítez [8] on cross-validation for temporally ordered data. We propose neither a better rating nor a better architecture: we measure the distance between a well-built supervised model and the market.

3 Materials

Sources and cleaning. Match results come from the TML-Database[1] (75,010 ATP matches, 2000–2024: rankings, surface, indoor flag, serve statistics,...). Odds come from tennis-data.co.uk (66,702 matches; two individual books plus the market average and maximum). We retain 65,281 matches (87.0%), excluding Davis Cup ties (7,081; team competition, different incentives), retirements, walkovers and defaults (2,436), and matches with a missing ranking (212). Retirements are excluded from training (the outcome does not reflect superiority on court) but remain in the history used to compute features, since a match played halfway is still informative about accumulated load. Walkovers never enter the history as played matches: neither player took the court.

Trivial baselines. The higher-ranked player wins 65.37% of matches; the favourite of the reference bookmaker wins 69.43%, and its mean overround of 1.0237 is a 2.37% margin that any betting strategy must clear. Predictability is context-dependent (best-of-five 71.7% vs best-of-three 63.9%), motivating the inclusion of match context among the features. Throughout, the market means the closing odds of the lowest-margin book among our four sources.

Record linkage. The two sources share no identifier and write names differently ("Juan Martin del Potro" vs "Del Potro J.M."), so we join on the player pair plus a $\pm$6-day window on the date. Compound surnames, frequent on tour, make the forename/surname boundary ambiguous: for every full name we generate all plausible segmentations and accept the join if any matches the abbreviated form, giving 60,082 of 65,281 matches (92.0%). Because the outcome is not part of the join key, the winner recorded independently by the two sources is a free integrity check.

4 Methods

Symmetric representation. In the raw data the winner is always listed first, so a feature built as "winner minus loser" would encode the label outright. We therefore assign the two players to slots A and B at random with a fixed seed and set $y = 1$ if A wins, which gives a class balance of 0.4971. Player attributes enter only as differences $d_\bullet = x_A - x_B$, which change sign when the slots are swapped, alongside context terms (surface, round, format) that do not change at all. One would expect predictions built on such features to satisfy $P(A \text{ beats } B) = 1 - P(B \text{ beats } A)$; they do not, because not everything flips: the differences do, but the surface, the round, and the model's own baseline are the same in either ordering, and that is what breaks the symmetry. We remove it at test time by predicting each match in both orderings and averaging:

$$p_{\text{sym}} = \tfrac{1}{2}\big[\, p(A, B) + 1 - p(B, A) \,\big].$$

The two predictions cancel term by term. The residue becomes exactly zero.

Causal features. All 38 features are produced in a single chronological pass in which, for each match, the accumulated history is read to form the feature vector and only afterwards updated with the outcome. Matches are ordered by tournament date, then tournament, then round, then match number, necessary because the recorded date is the tournament start date, identical for every match of an event. The blocks are: Elo ratings; log-transformed ATP rank and points (the rank–outcome relation saturates beyond $\sim$100 places); recent form over 10/25/50-match windows and surface-specific form; load and rest; injury proxies; technical statistics averaged over the previous 20 matches; head-to-head; demographics; and match context. Leak-freeness is verified by 14 automated tests. The decisive one flips the outcome of a match in the middle of the dataset and asserts that no feature of any earlier match changes; a second checks exact antisymmetry of every $d_\bullet$; a third rejects any feature correlating above 0.6 with the outcome, a typical leakage signature (the maximum is 0.42).

[1] `https://github.com/Tennismylife/TML-Database`

Elo. Two parallel rating systems, one global and one per surface, are combined linearly with weight λ; the K-factor decays with experience as $k_0/(n+\text{offset})^{\text{shape}}$, so newcomers move fast and veterans are stable, with parameters chosen by grid search on the training period only. The robust finding is not the value of k_0 or the decay exponent but the surface weight: $\lambda = 0.5$ wins for every combination of the other parameters, and surface-only Elo is systematically worse than global Elo. Specialisation fragments the data more than it informs.

Validation design. We use rolling-origin cross-validation with 32 quarterly folds over 2011–2018, the training window expanding over all history from 2003. The years 2019–2024 are held out from every choice: the quality comparison evaluates on all six, while the betting simulation retrains through 2022 and bets on 2023–2024 only. The reference model is compared with each competitor through the paired differences over the 32 folds, reported as a mean with a 95% Student confidence interval, $\bar{d} \pm t_{31,\,0.975}\, s_d/\sqrt{n}$. The question here is how large the differences are, not whether they are exactly zero, and an interval answers that directly: it places all 11 comparisons on one continuous scale, and it stays valid under the selection of the reference. Picking the best model from the same folds inflates the measured differences, so an upper bound is if anything conservative.

Models and protocol. We compare 11 classifiers: logistic regression (ℓ_1, ℓ_2), Gaussian naive Bayes, k-NN, decision tree, random forest, gradient boosting, AdaBoost, a feed-forward network, a linear SVM, and an RBF SVM. The kernel SVM solves a quadratic problem on an $n \times n$ Gram matrix, prohibitive at $n \approx 40\,000$, so it is trained on a random 10 000-match subsample of each training fold: the kernel is exact, the restriction is on the data and is chosen as a computational budget rather than by accuracy. All models run on scikit-learn [9] with standard, non-tuned configurations, fixed once and never searched over: tuning each to a different degree would confound algorithm quality with search effort. Numeric features are median-imputed and standardised, categorical ones one-hot encoded, all fitted within each training fold; the configuration used throughout is gradient boosting with test-time symmetrisation. A bet is placed when $p_{\text{model}} > 1/o$; we use flat stakes across all four odds sources and a range of edge thresholds, with bootstrap confidence intervals on the return.

5 Results

The algorithm does not matter; the representation does. Table 1 shows five models converging to the same value in the fourth decimal place. Two penalised logistic regressions, a maximum-margin linear separator and two tree ensembles cannot be told apart: whatever their true differences are, the paired intervals bound them above by 0.0021, under a quarter of the 0.0093 that separates all five from Elo, whose own interval is $[0.0068,\ 0.0117]$. This is not the absence of evidence for a difference but evidence that any difference is too small to matter, and it locates the ceiling in the representation rather than in the hypothesis class. Note that per-fold standard deviations (± 0.024) dwarf the spread between the top five models (0.0010): variation across quarters is far larger than variation across models, but it is common to all of them and cancels in the paired difference, which is why intervals this tight are attainable at all.

Table 1: Classifier comparison, 32 quarterly rolling-origin folds (2011–2018). Elo is not trained and serves as reference. For every model above the rule the 95% interval on the paired difference from the best contains zero and is bounded above by 0.0021; for every model below it the interval excludes zero.

Model	Accuracy	Log-loss
Gradient boosting	69.51%	**0.5771**
Logistic (ℓ_1)	69.33%	0.5775
Linear SVM	69.26%	0.5776
Logistic (ℓ_2)	69.25%	0.5777
Random forest	69.22%	0.5781
Feed-forward NN (64,32)	69.24%	0.5813
k-NN ($k = 100$)	68.81%	0.5855
Elo (baseline)	*68.92%*	*0.5864*
Decision tree	68.53%	0.5932
AdaBoost	68.95%	0.5946
SVM RBF	69.29%	0.5972
Naive Bayes	68.44%	2.0265

Accuracy and log-loss do not rank models alike. The RBF SVM is third by accuracy (69.29%) and second from last by log-loss (0.5972); AdaBoost reaches 68.95% with a log-loss worse than Elo's; naive Bayes reaches a respectable

68.44% with a log-loss of 2.03. All three discriminate well and quantify badly. Naive Bayes collapses because its independence assumption is flagrantly violated: the three Elo features are exactly collinear by construction (the blend is the mean of the other two), so the same evidence is counted three times and the posterior is driven to the extremes; AdaBoost suffers from the exponential loss for a structurally similar reason, and the RBF SVM returns a margin whose conversion to a probability is fitted on a subsample. For the betting application, where the probability and not the label is what matters, all three are unusable despite their accuracy.

What the features actually contribute. Six blocks of nine have an interval excluding zero (Table 2), but for two of them (ATP ranking and technical statistics) the lower bound is $+0.0001$, so the evidence is thin and the interval says so directly. Redundancy is visible in their profile: ATP ranking alone reaches 66.84% accuracy, far above its marginal contribution of 0.0014, because Elo has already absorbed most of what it carries. Head-to-head contributes nothing (its interval, $[-0.0005, +0.0006]$, brackets zero so tightly that any contribution is at most 0.0006) despite 57.09% standalone accuracy: the information is already in Elo and recent form, and the exploratory analysis shows why: of 35,368 distinct pairs, 62.9% have met exactly once. Demographics are the opposite case: age, height and handedness reach 51.81% alone, barely above chance, yet removing them costs 0.0025 with an interval well clear of zero. They predict nothing on their own but modulate the other features, a purely interactive contribution, invisible to univariate screening. Match context alone gives 49.96%, chance to two decimal places, which independently validates the symmetric representation: surface, round and format cannot say who wins, only how predictable the match is, and any departure from 50% would have signalled a construction defect.

Table 2: Feature-block ablation. *Damage* is the log-loss increase when the block is removed from the full model (0.5771); *alone* is the log-loss using that block only. Marginal contribution and standalone power measure different things, and reading either alone misleads.

Block	Damage	95% CI	Alone	Acc. alone
Elo	$+0.0049$	$[+0.0032, +0.0065]$	0.5869	68.71%
Demographics	$+0.0025$	$[+0.0014, +0.0036]$	0.6918	51.81%
Recent form	$+0.0020$	$[+0.0007, +0.0033]$	0.5964	67.94%
ATP ranking	$+0.0014$	$[+0.0001, +0.0028]$	0.6050	66.84%
Technical stats	$+0.0013$	$[+0.0001, +0.0025]$	0.6216	65.24%
Load and rest	$+0.0013$	$[+0.0005, +0.0021]$	0.6496	62.29%
Match context	$+0.0006$	$[-0.0000, +0.0013]$	0.6934	49.96%
Injury proxies	$+0.0004$	$[-0.0004, +0.0012]$	0.6813	56.10%
Head-to-head	$+0.0000$	$[-0.0005, +0.0006]$	0.6698	57.09%

The model beats Elo and loses to the market. On the untouched 2019–2024 hold-out (14,360 matches never used for model selection), over the 12,824 of them for which market odds exist, Elo scores 64.50% accuracy and 0.6248 log-loss, our model 65.86% and 0.6076, the market 68.08% and 0.5848: the bookmaker's edge over us (0.0227) is larger than our edge over Elo (0.0172). On the 2023–2024 betting test the ordering is identical (66.04% vs 68.29%), and the return is negative for every threshold and every odds source, with bootstrap intervals excluding zero; the best case is -8.45%. One diagnostic makes this interpretable: the return worsens as the edge threshold rises, in all four sources (Fig. 1). Were the measured edge real information, concentrating on the largest edges would improve the return; that it deteriorates identifies the "edge" as model error: the model bets predominantly on underdogs and systematically overvalues them. A sanity check validates the pipeline rather than the model: betting systematically on the favourite returns approximately minus the margin at every source (-4.45% at a 5.17% margin, -1.76% at 2.62%, $+0.14\%$ at 0.54%), whereas an error in orienting odds from winner/loser to A/B would have produced implausible returns.

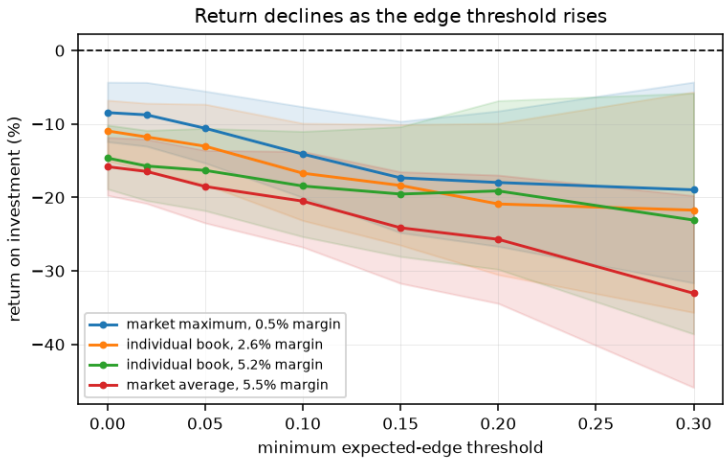


Figure 1: Return on investment against the minimum edge threshold, by odds source, with bootstrap confidence bands. The decline as the threshold rises is the diagnostic signature of a spurious edge.

Where the gap lives. The market wins in all 29 segments examined, but the magnitude is systematic. The gap is smallest where history is dense ($+0.0087$ for pairs with more than two previous meetings, $+0.0087$ when both players have over 300 career matches) and almost five times larger where it is thin: $+0.0407$ when at least one player has fewer than 20 career matches. This is an *information* deficit, not a modelling one, and it is precisely where a bookmaker's scouting and first-hand knowledge have no counterpart in result data.

6 Conclusion

Under a symmetric, verifiably causal representation, five classifiers of very different inductive bias converge to the same performance, all clearly above a tuned surface-blended Elo and all clearly below the betting market. The binding constraint is therefore the information in result data: the model–market gap is narrowest where match history is dense and widens almost fivefold where it is sparse. The remedy is not a better classifier but sources carrying current physical condition and information on players with thin history.

Limitations. Odds are closing odds and the simulation ignores stake limits and commissions: it measures informational content, not a tradable strategy. Hyperparameters were fixed once and never searched over, deliberately, to keep the comparison between families fair; a tuned model would likely gain a little, though the convergence of five model classes suggests very little.

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